Persistence Mechanics of the Quantum Vacuum I: The E_8 Substrate and the Fine-Structure Constant

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Persistence Mechanics (PM) holds that any system that persists must satisfy Finiteness, Conservation, and Causality, realized through six pillars: Capacity, Map, Protocol, Governor, Toll, and Margin. We apply it to the quantum vacuum, treated as a closed system with zero external flux. Writing PM's requirements as explicit postulates on a discrete substrate, we prove in Lean 4 that they admit exactly one substrate up to isometry: the E_8 lattice, with two orthogonal D_4 sectors and a 43-state clock. Every integer that enters the cost of persistence is then forced. Reading each pillar as an expected number of transitions per cycle gives the structural index $Z_{\text{geo}} = 43\pi + 2 - \frac{1}{167} - \frac{2}{43} + \frac{334}{32^3 \cdot 5 \cdot 172} + \frac{1}{31 \cdot 6 \cdot 43^2} = 137.035\,999\,212\ldots$, certified to eighteen decimal places, with no continuous parameter. Under an electromagnetic matching condition $Z_{\text{geo}} = \alpha^{-1}(0)$, it agrees with Morel et al. (2020) within $0.58\,\sigma$ and CODATA 2022 within $1.68\,\sigma$, and lies $3.1\,\sigma$ and $6.2\,\sigma$ from the Fan (2023) and Parker (2018) determinations. The proofs use only Lean's standard axioms; their premises are the stated postulates and PM's definitions of the pillars. The matching condition is a separate physical hypothesis.

I. INTRODUCTION

The idea that physical reality is constrained by the cost of processing information goes back to Landauer's principle[1] and Wheeler's "It from Bit"[2]. Verlinde[3] and Bianconi[4] have proposed that gravity and spacetime emerge from information gradients. These programs motivate a systems-first question: if physical structure is informationally constrained, what minimal architecture makes that information processing possible at all? Causal sets[5], loop quantum gravity[6], and emergent gravity[3, 7] start from spacetime or gravity. Here we start from persistence.

Persistence Mechanics[8] (PM) describes what any system must instantiate to remain identifiable through time. Its three requirements are Finiteness (bounded resources), Conservation (preserved identity-defining distinctions), and Causality (well-ordered evolution). They are realized by six pillars: Capacity and Margin (Finiteness), Map and Protocol (Conservation), Toll and Governor (Causality). Operating the pillars costs transitions, and histories that cost more are exponentially less likely to survive.

PM is usually applied to systems that persist inside an environment: cells, networks, organizations. We ask how far down it reaches. The quantum vacuum is the limiting case: a persistent system with zero external flux, no deeper substrate, and nothing outside it to supply memory, a clock, or a unit of measure. If PM's requirements, applied to such a system, single out its structure, the vacuum is the baseline instance of persistence. If they do not, the vacuum's structure is set by something PM does not describe.

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A. Result

We formalize PM's requirements for the vacuum as explicit postulates on a discrete substrate (Table I) and prove the consequences in Lean 4. The postulates admit exactly one substrate up to isometry, the E_8 lattice (Section III). It carries two orthogonal D_4 sectors, 16 channel coordinates per sector, one time direction, a spherical defect boundary, and a clock with 43 states (Section IV). Every integer that enters the cost of persistence is forced.

Each pillar's cost is an expected number of transitions per cycle. On the forced substrate the six terms sum to the structural index (Section V)

$$Z_{\text{geo}} = 43\pi + 2 - \frac{1}{167} - \frac{2}{43} + \frac{334}{32^3 \cdot 5 \cdot 172} + \frac{1}{31 \cdot 6 \cdot 43^2}. \quad (1)$$

Lean certifies $137.035999212381397211 < Z_{\text{geo}} < 137.035999212381397213$. No continuous parameter enters. Under an electromagnetic matching condition, Z_{geo} is the inverse fine-structure constant $\alpha^{-1}(0)$; we compare it with measurement and draw the consequences in Section VI.

B. What is proved, and from what

The main theorem (Lean: `Main.persistence_selects`) states: from the postulates of Table I, the substrate is E_8 with two D_4 sectors, the clock period is 43, the parameter record is forced, and Z_{geo} is given by Equation (1) and lies in the certified interval. A second theorem shows that E_8, the 43-cycle, one time direction, and the sphere satisfy every postulate, so the set is consistent. Each geometric postulate is independent of the others: for each, Lean exhibits a lattice that meets every other postulate but not that one.

The premises are the postulates themselves and PM's definitions of the pillars, in particular that Governor restricts transitions and Protocol coordinates them. The proofs use only Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`); a strict audit checks every one of the project's 1236 declarations. The electromagnetic matching condition of Section VI is not part of the theorem. The Lean project, with a map from each claim to its theorem, is in the `lean/` directory of the accompanying repository.

C. Scope

This paper treats the static, quiescent baseline of the vacuum. Dynamics, a Lagrangian, gauge structure, a chiral matter spectrum, and Lorentz invariance of the continuum limit are extensions, not prerequisites for the question asked here. In particular, the Distler–Garibaldi analysis of E_8 embeddings[9] constrains any later derivation of chiral matter; we state what such a derivation must establish in Appendix A.

II. THE VACUUM AS A CLOSED PERSISTENT SYSTEM

We apply PM to the vacuum in its quiescent steady state: stationary operation at zero external flux. The vacuum has no deeper system beneath it and no environment beside it. It cannot obtain configuration data, boundary conditions, synchronization signals, a measuring probe, or a unit of cost from outside itself. We state this as one constraint.

> **Closure Constraint.** The vacuum's state space, its update, and everything used to read, time, or price its states belong to the vacuum itself.

Closure is used in three concrete ways below.

(i) **Internal probes.** Any readout of the substrate is made by the substrate. A probe is itself a substrate state, and its outcome is held at the substrate's own resolution (Section III B).

(ii) **No preferred label.** Nothing outside can single out one element of a finite set of labels the substrate cannot distinguish internally. Admissible sets are then empty or everything, and the only admissible probability on such a set is uniform (Lean: `ClosureSymmetry.invariant_set`, `invariant_measure_uniform`).

(iii) **One currency.** There is no external unit. Lengths, counts, and probabilities are all measured in elementary transitions, with one transition as the unit of cost.

Conservation is read in PM's sense: the substrate preserves the distinctions that define its identity. On a finite state space with an autonomous update this means the update never merges two states, which is exactly bijectivity (Lean: `never_merge_iff_bijective`).

III. SYSTEM 0: THE SUBSTRATE POSTULATES

This section states the postulates that select the substrate and derives their consequences. Each postulate is a formal reading of a PM requirement for a closed system. None names E_8, a rank, or a target number.

A. Addresses and Cost

Postulate L. Microscopic addresses form a full-rank lattice $\Lambda \cong \mathbb{Z}^n$ whose translations preserve the local update, with an integral symmetric configuration cost $B(x, y) = x^{\mathsf{T}} A y$ and $Q(x) = B(x, x)$. The cost is integral because, by Closure, it counts transitions.

Postulate F (Finiteness). For every c, the set $\{x \in \Lambda : Q(x) \le c\}$ is finite.

For an integral symmetric form, F holds if and only if A is positive definite (Lean: `posDef_iff_finite_levels`). An indefinite cost would allow infinitely many configurations at bounded cost.

B. Readability and Self-Duality

By Closure a probe is a substrate state y, and reading a state x with it yields the pairing $B(x, y)$. The smallest distinction is one bit.

Postulate R (Conservation under Closure). Every one-bit distinction of the substrate is readable by the substrate at one-bit resolution: if $B(x, y)$ is even for every probe $y \in \Lambda$, then $x \in 2\Lambda$.

Theorem 1. *Postulate R holds if and only if* $\det A$ *is odd.*

The states invisible to every probe at one-bit resolution form $(\Lambda \cap 2\Lambda^*)/2\Lambda$, the 2-torsion of the discriminant group Λ^*/Λ (Lean: `bitReadable_iff_odd`). An undetectable label would be a distinction the closed substrate carries but cannot maintain, since nothing inside it can read it.

Postulate A (Map). Addresses are read as bits without an external translation table: every glue element is its own inverse, $2\Lambda^* \subseteq \Lambda$.

Theorem 2. *Postulates R and A together imply* $\det A = \pm 1$*: the substrate is unimodular,* $\Lambda = \Lambda^*$.

By A, Λ^*/Λ is 2-torsion; by R it has no 2-torsion; so it is trivial (Lean:

Postulate	PM requirement	Lean	Consequence
L. Addresses form a lattice with integral symmetric cost	Finiteness	`Vacuum.Substrate`	Gram matrix $A \in M_n(\mathbb{Z})$
F. Bounded-cost sets are finite	Finiteness	`Geometric.finite`	positive definite
R. One-bit distinctions are readable by the substrate	Conservation, Closure	`Geometric.readable`	$\det A$ odd
A. Glue elements are bits, $2\Lambda^* \subseteq \Lambda$	Map	`Geometric.bitAddressed`	with R: unimodular
X. Nontrivial exchange; zero pairing commutes	Protocol, Causality	`Geometric.exchange`	even
M. Least rank among such substrates	Economy (selection)	`Vacuum.Economical`	rank 8, E_8
B. Two orthogonal irreducible bit-addressed rank-4 root sectors	Map, Protocol	`Vacuum.SectorPostulates`	$D_4 \oplus D_4$, $\nu = 16$, $N = 32$
C. Clock is a substrate plane; states = its discriminant group; unique geometry	Closure, Map	`ClockPostulates.plane`	complex multiplication, $\Delta = \lvert D \rvert$
I. Schedule reconstructable from gap-only local observations	Causality	`ClockPostulates.schedule`	$32 \le \Delta \le 64$, so $\Delta = 43$
G. Coordinate volume element squares to -1	Map	`SignaturePostulate`	one time direction
H. Defect boundary of least topological complexity	Economy	`BoundaryPostulate`	sphere, $\chi = 2$

Table I. The postulates. Each formalizes a PM requirement for the closed vacuum; the last column is what Lean derives from it. Postulates F–M select the substrate (Section III); B–H fix its internal structure (Section IV).

`bitAddressed_bitReadable_unimodular`). Equivalently, $\det A \cdot \det(2A^{-1}) = 2^n$ with $\det A$ odd.

Self-duality is thus not stated directly. It follows from two requirements about how a closed system reads its own addresses. Each alone is weaker: D_4 satisfies A but not R, and A_2 satisfies R but not A (Table II).

C. Exchange and Evenness

Translation operations U_x act on the substrate and compose up to a central phase. Their exchange law $U_x U_y = \varepsilon^{C(x,y)} U_y U_x$, with ε of order two, defines a bi-additive form C with values in $\mathbb{Z}/2$. An operation always commutes with itself, so $C(x,x) = 0$.

Postulate X (Protocol and Causality). Some pair of operations fails to commute ($C \neq 0$): the order of operations is observable somewhere. Operations whose addresses have zero pairing commute: coordination requires a coupling channel.

Theorem 3. *Postulate X implies that every norm $Q(x)$ is even.*

If $Q(v)$ is odd, then $w = Q(v)z - B(v,z)v$ is orthogonal to v for every z, so locality gives $Q(v)C(v,z) = C(v,w) = 0$ and hence $C(v,\cdot) = 0$. Additivity of norm parity spreads this to every row, so $C = 0$, contradicting nontriviality (Lean: `orthogonal_nontrivial_exchange_forces_even`). Locality is what excludes the parity-corrected exchange law $B(x,y) - Q(x)Q(y) \bmod 2$, which is consistent on odd lattices but makes orthogonal operations anticommute.

D. Economy and the Selection of E_8

Postulates F, R, A, X say exactly that A is positive definite, even, unimodular, and nonzero:

Theorem 4 (Equivalence)**.** *A substrate satisfies F, R, A, and X if and only if its lattice is positive definite, even, unimodular, and of positive rank.*

(Lean: `Vacuum.geometric_iff`.) The converse direction supplies the witnesses: $2 \det A \cdot \operatorname{adj} A$ realizes A, and the reduced pairing realizes X.

Postulate M (Economy). No substrate satisfying F, R, A, and X has smaller rank. This is PM's selection principle: cost grows with the number of independently maintained coordinates, and surviving histories concentrate on least cost.

Theorem 5. *A nonzero positive-definite even unimodular lattice has rank divisible by eight.*

The proof is the theta-series argument[10, 11]. Evenness gives $\Theta(s - i) = \Theta(s)$; unimodularity and Poisson summation give $\Theta(1/s) = s^{n/2}\Theta(s)$; following $1 \to 1+i \to (1+i)/2 \to 1$ yields $\Theta(1) = x^n \Theta(1)$ with $x^2 = -i$, and $\Theta(1) > 0$ forces $8 \mid n$ (Lean: `rank_dvd_eight`, with n-dimensional Poisson summation proved by induction on the dimension).

Lattice	F	R	A	X	M
hyperbolic plane	×	✓	✓	✓	
D_4	✓	×	✓	✓	
A_2	✓	✓	×	✓	
$\mathbb{Z}$	✓	✓	✓	×	
$E_8 \oplus E_8$	✓	✓	✓	✓	×

Table II. Independence. Each lattice satisfies every postulate but one (Lean: `finiteness_needed`, `readability_needed`, `bitAddressing_needed`, `exchange_needed`, `economy_needed`).

Theorem 6. *Up to integral isometry, the only positive-definite even unimodular lattice of rank eight is E_8.*

The theta series is a modular form of weight four for $SL_2(\mathbb{Z})$, hence equal to the Eisenstein series E_4; this gives 240 roots and 2160 vectors of norm four. A frame of eight orthogonal norm-four vectors congruent modulo 2Λ spans a D_8, and evenness with unimodularity fixes the glue to E_8 (Lean: `thetaMF_eq_E4`, `card_norm_four`, `e8_unique`).

Since E_8 satisfies F, R, A, and X, Economy bounds the rank by eight; divisibility forces rank eight; uniqueness gives E_8 (Lean: `Vacuum.geometric_selects_e8`). Conversely E_8 is economical, so the selection is consistent and unique (`substrate_exists_unique`). Its packing optimality[12] is not used.

IV. SYSTEM 1: STRUCTURE OF THE SUBSTRATE

With the substrate fixed as E_8, the remaining postulates of Table I fix its internal structure: sectors, channels, signature, boundary, and clock. Together they force every integer used in System 2 (Lean: `Derived.parameters_forced`).

A. Sectors

Postulate B. The substrate contains, with finite index, two orthogonal rank-four sectors of the same type, one for addresses and one for internal state. Each is generated by norm-two vectors, is irreducible (not itself an orthogonal sum), and is bit-addressed ($2\Lambda_s^* \subseteq \Lambda_s$, Postulate A applied to a sector).

A positive-definite lattice generated by norm-two vectors is a simply-laced root lattice, so the rank-four candidates are A_4, D_4, A_3A_1, A_2A_2, $A_2A_1A_1$, and A_1^4. Only D_4 is irreducible with involutive glue: A_4 has glue group $\mathbb{Z}/5$ and A_1^4 is reducible (Lean: `sector_selection`). E_8 contains two orthogonal D_4 sectors, given by an explicit basis whose Gram matrix is $D_4 \oplus D_4$ (`e8_two_d4_sectors`); their sum has index four, and its four glue cosets reconstruct E_8. The address sector has rank $D = 4$.

B. Channels

A sector's glue group $D_4^*/D_4 \cong (\mathbb{Z}/2)^2$ has three nonzero classes. We define a *channel* as a nonzero glue class, and its capacity as its number of minimal (least-cost) vectors. Each of the three classes has exactly eight (Lean: `minimal_class_card`); these are the classes usually labeled 8_v, 8_s, 8_c and permuted by triality[13].

Directed flow needs a source and a sink, so at least two channels (Causality); Economy allows no more. Hence each sector carries

$$\nu = 2 \times 8 = 16 \tag{2}$$

channel coordinates, whichever two classes are used, and the node carries $N = 2\nu = 32$ (`channel_total`). The third class carries no flow. Pure lattice symmetry would treat all three classes alike; the two-channel rule is where Causality breaks triality.

C. Signature and Boundary

For a four-dimensional orthogonal Clifford basis with q negative squares, the volume element obeys $\omega^2 = (-1)^q$ (Lean: `signature_volume_square`).

Postulate G. The volume element of the address sector supplies its complex structure, $\omega^2 = -1$; with the overall sign convention, time directions are the minority.

Then $q = 1$: one time direction and three spatial ones (`one_time`). The internal configuration cost remains positive definite; the Lorentzian signature belongs to the effective coordinate sector.

Postulate H. A persistent defect is bounded by a closed connected surface of no greater topological complexity than any other, that is, of maximal Euler characteristic.

Among closed surfaces $\chi \leq 2$, with equality only for the sphere (Lean: `euler_eq_two_iff`, `boundary_sphere`). So the boundary is S^2 and

$$\chi = 2. \tag{3}$$

A defect's configuration is a spatial position and a boundary point, of dimension

$$\sigma = (D - q) + 2 = 3 + 2 = 5. \tag{4}$$

D. The Clock

Postulate C. By Closure the clock is built from the substrate: its period lattice is a nondegenerate plane $\Lambda_c = \mathbb{Z}u + \mathbb{Z}v \subset \Lambda$. Its states are the plane's discriminant group Λ_c^*/Λ_c, the finite group determined by the plane and its form alone, and a tick advances through all of them. The clock geometry is determined by its data: all clock tori of the same discriminant are isomorphic.

Because E_8 is even, the half-norm $\frac{1}{2}Q(xu+yv) = ax^2+bxy+cy^2$ is an integral binary form of discriminant $D_c = B(u,v)^2-Q(u)Q(v)$ (Lean: `plane_form`). The shape $\tau = v/u$ satisfies $a\tau^2+b\tau+c=0$, so the clock torus $\mathbb{C}/\Lambda_c$ has complex multiplication; this is forced by evenness, not assumed. The discriminant group has exactly $|\det G_c| = |D_c|$ elements for any nondegenerate Gram matrix G_c (Lean: `card_discriminant_group`), so the period is

$$\Delta = |D_c|. \tag{5}$$

Tori of discriminant D_c correspond to $SL_2(\mathbb{Z})$-classes of forms[14], so a unique clock geometry means class number one. Over all imaginary quadratic orders this occurs exactly for[15]

$$|D_c| \in \{3,4,7,8,11,12,16,19,27,28,43,67,163\}. \tag{6}$$

Postulate I. Each cycle serializes the N channels as active slots, each followed by an idle run. Runs are uniformly bounded, and the schedule is reconstructable from local observations that record only whether an idle run follows a transition, with no external clock or run-length counter.

Gap-only observations reconstruct schedules with runs of length at most k if and only if $k \le 1$ (Lean: `obs_injOn_iff`); longer runs alias (`aliasing`). Hence

$$N \le \Delta \le 2N, \qquad 32 \le \Delta \le 64 \tag{7}$$

(`period_window`). By Gauss reduction, the only discriminant in this window with a unique positive form is -43, proved directly without the classification (`clock_unique_iff`); the vendored class-number-one classification for number fields gives the same window independently. Therefore

$$\Delta = 43. \tag{8}$$

E_8 contains such a plane: $Q(u) = 2$, $B(u,v) = 1$, $Q(v) = 22$, half-norm form $x^2+xy+11y^2$, discriminant group $\mathbb{Z}/43$ visited in a single cycle (`e8_clock_plane`, `clock_glue`). A schedule with eleven idle slots realizes the bound (`e8_clock`). Combining the sections so far, the postulates force E_8, two D_4 sectors, and $\Delta = 43$ (`Vacuum.vacuum_selects`).

E. Derived Quantities

a. Addresses per cycle. Each of the D coordinates is addressed at each of the Δ phases:

$$R_M = D\Delta = 172. \tag{9}$$

b. Description length. Over one transition a defect is specified by its channel content (ν coordinates), its configuration (σ), its boundary closures (χ), and its current and next address (D each, Causality). A description that keeps only some of a state's independent coordinates distinguishes all states exactly when it keeps every one (Lean: `description_minimal`); Conservation requires all of them and Economy forbids more. The description therefore has

$$L_{\text{embed}} = \nu+\sigma+\chi+2D = 31 \tag{10}$$

coordinates. The substrate gives the same number intrinsically: E_8 has 240 roots and rank 8, and $(240+8)/8 = 31$ is its count of nearest-neighbour moves plus coordinate directions per coordinate, the dimension of $\mathfrak{e}_8$ per rank (`description_length`).

All integer inputs are now fixed: $D = 4$, $\Delta = 43$, $\nu = 16$, $N = 32$, $\sigma = 5$, $\chi = 2$, $R_M = 172$, $L_{\text{embed}} = 31$.

V. SYSTEM 2: THE COST OF PERSISTENCE

We now price the persistence of a minimal defect on the forced substrate. Each of the six pillars contributes one term; each term is computed from the integers of System 1.

A. Costs as Expected Transitions

PM prices a pillar as toll times frequency, $Z_i = c_0 f_i$, sums the pillars, and only then exponentiates: a history of action S survives with probability $e^{-\beta S}$[8]. Three consequences fix how the terms combine.

(i) **Each term is an expected count.** A pillar's cost per cycle is the expected number of transitions it accounts for. For indicator events, expected counts add, each weighted by the single toll c_0 (Lean: `expected_count_linear`); Closure sets $c_0 = 1$. A probability therefore enters as itself, not as $-\log p$, and the terms add with unit weight. The exponential applies to the total: survival is the product of per-pillar factors $e^{-\beta Z_i}$ (`survival_factorizes`).

(ii) **Signs follow the pillar definitions.** Capacity, Margin, and Map provision, and Toll spends; their transitions add. Governor restricts which transitions occur and Protocol coordinates them; both act on other pillars' transitions and create none. Restriction keeps a subset and coordination merges equivalent attempts, so neither can increase the count, and a removed transition lowers it by exactly its probability (`restriction_lowers`, `merging_lowers`, `removal_subtracts`). Each of Conservation and Causality has one regulating pillar; Finiteness has none (`NetIndex.signs`).

(iii) **Uniform measures.** Where a term is a probability over a finite label set that the substrate cannot distinguish internally, Closure forces the uniform measure (Section II).

The index is therefore the net number of transitions per cycle, provisioned minus removed (`index_as_net_transitions`):

$$Z_{\text{geo}} = Z_{\text{CAP}} + Z_{\text{MAP}} + Z_{\text{TOL}} + Z_{\text{MAR}} - |Z_{\text{PRO}}| - |Z_{\text{GOV}}|. \tag{11}$$

B. Capacity

A defect's identity is its set of distinctions from the vacuum. Let the maintenance cost be local, measured from the vacuum, and unchanged by exchanging the names of the two states (by Closure, which state is called "vacuum" is not intrinsic). Then the cost is one density times the number of neighbouring pairs in different states (Lean: `only_distinctions_cost`). Uniform regions, the defect's interior and the surrounding vacuum, cost nothing; only the boundary costs. An area cost satisfies the first two conditions but not the third (`area_breaks_relabeling`). At speed one, one unit of boundary is one tick, so the density is one transition.

The defect's envelope lies in the causal plane. A signal must cross it and return within one cycle, so its width in every direction is at most Δ. For a convex envelope with support function h, the width in direction θ is $h(\theta)+h(\theta+\pi)$ and the perimeter is $\int_0^{2\pi} h$. Width at most w in every direction gives perimeter at most πw, with equality for every envelope of constant width w, whatever its shape (Barbier's theorem; Lean: `perimeter_le`, `barbier`). Capacity is the allocation the causal bound permits:

$$Z_{\text{CAP}} = \pi\Delta \approx 135.088\,484\,104\,361 \tag{12}$$

(`capacity_is_max`). Neither roundness nor saturation is a separate assumption.

C. Map

Map retains the boundary that defines the defect. For any closed triangulated surface, the total angle defect is $2\pi(V - E + F) = 2\pi\chi$, independent of triangulation, shape, and size (Descartes' theorem, the discrete Gauss–Bonnet theorem; Lean: `descartes`). So χ is the number of full 2π turns the boundary makes to close: two for the sphere (`sphere_two_turns`). Counting one transition per closure,

$$Z_{\text{MAP}} = \chi = 2. \tag{13}$$

D. Protocol

The defect's embedding occupies σ of the R_M addresses of a cycle; any injective placement leaves exactly $R_M - \sigma = 167$ residual addresses (Lean: `residual_card`). The update is a function, so a state has exactly one successor. Protocol coordinates that successor among the residual addresses without an external pointer; the coordination event has the uniform probability of a single address, 2^{-I} with $I = \log_2 167$ the information needed to name it (`protocol_persistence`). It removes one transition when it occurs:

$$Z_{\text{PRO}} = -\frac{1}{R_M - \sigma} = -\frac{1}{167} \approx -0.005\,988\,023\,952. \tag{14}$$

E. Governor

Governor restricts transitions using the boundary's turning budget of χ closures per cycle. No slot of the cycle is preferred, so a credit distributed over the Δ slots that is invariant under advancing the clock is uniform, with χ/Δ per slot (Lean: `governor_uniform`). It removes transitions:

$$Z_{\text{GOV}} = -\frac{\chi}{\Delta} = -\frac{2}{43} \approx -0.046\,511\,627\,907. \tag{15}$$

F. Toll

Toll is the probability that a transition preserves the defect.

a. Three legs. A leg's state is what the substrate observes locally about its channel: active or idle, one bit (Postulate I; Lean: `leg_state_binary`, `leg_factorization`). By Closure the legs carry no external labels, so an interaction treats them symmetrically; by Conservation it is reversible. Every such two-leg update relabels each leg by the same map, possibly exchanging them, so no leg depends on the other (`two_legs_separable`). A symmetric reversible three-leg update exists that couples the legs (`three_legs_interact`). Economy takes the least arity that coordinates: three. (With three-valued legs two would suffice, `three_valued_two_legs_interact`, so the one-bit leg is essential.)

b. Valid transitions. The internal configuration of a vertex is its three legs, each one of N channels, and its interaction direction, one of σ. A transition preserves the defect when the configuration returns to the reference up to the defect's one invisible symmetry, reversal of boundary orientation. A connected orientable boundary has exactly two orientations (`two_orientations`), related only by the flip (`orientation_relabelings`), so exactly two configurations are valid (`toll_identity`). The target address must also avoid the defect's own embedding (Causality: no self-aliasing), which a uniform target does with probability $(R_M - \sigma)/R_M$ (`nonaliasing_prob`). Under the uniform measure,

$$Z_{\text{TOL}} = \frac{2}{N^3\sigma} \cdot \frac{R_M - \sigma}{R_M} = \frac{334}{32^3 \cdot 5 \cdot 172} \approx 1.185\,217\,569\,041 \times 10^{-5} \tag{16}$$

Pillar	Term	Value	Meaning
Capacity	$\pi\Delta$	$+135.088\,484\,104\,361$	largest causal boundary
Map	χ	$+2$	boundary closures
Toll	$\frac{2(R_M-\sigma)}{N^3\sigma R_M}$	$+1.185\,217\,569\,041\times10^{-5}$	persisting transition
Margin	$\frac{1}{L_{\text{embed}}(\sigma+1)\Delta^2}$	$+2.907\,703\,670\,104\times10^{-6}$	one configuration
Protocol	$-\frac{1}{R_M-\sigma}$	$-0.005\,988\,023\,952$	coordinated successor
Governor	$-\frac{\chi}{\Delta}$	$-0.046\,511\,627\,907$	restricted slots
Total	Z_{geo}	$137.035\,999\,212\ldots$	

Table III. The six terms on the forced substrate.

(`toll_persistence`). The factor 2 is the number of orientations; it equals χ only because the boundary is a sphere.

G. Margin

Margin is the resolution floor: one configuration among all the configurations of a localized defect. These are the product of three independent domains.

- **Description:** $L_{\text{embed}} = 31$ coordinates (Paragraph IV E 0 b).
- **Local interaction:** a connected subgraph on σ edges localized to one neighbourhood is a tree, and any tree with σ edges has $\sigma+1=6$ vertices; radius one makes it the star (Lean: `tree_vertices`, `radius_one_star`).
- **Causal cone:** at speed one in the causal plane, tick t of a cycle reaches $2t+1$ positions, so one cycle's causal cone holds $\sum_{t<\Delta}(2t+1)=\Delta^2$ events (`causal_cone_card`).

Under the uniform measure,

$$Z_{\text{MAR}} = \frac{1}{L_{\text{embed}}(\sigma+1)\Delta^2} = \frac{1}{31\cdot 6\cdot 43^2} \approx 2.907\,703\,670\,104\times10^{-6}. \tag{17}$$

H. The Index

Summing the six terms (Table III),

$$Z_{\text{geo}} = \pi\Delta + \chi - \frac{1}{R_M-\sigma} - \frac{\chi}{\Delta} + \frac{2(R_M-\sigma)}{N^3\sigma R_M} + \frac{1}{L_{\text{embed}}(\sigma+1)\Delta^2}. \tag{18}$$

Theorem 7 (Main theorem)**.** *Under the postulates of Table I, the substrate is E_8 with two D_4 sectors, $\Delta=43$, all integer inputs take the values of System 1, and*

$$137.035999212381397211 < Z_{\text{geo}},$$
$$Z_{\text{geo}} < 137.035999212381397213.$$

The postulates are jointly satisfiable.

(Lean: `Main.persistence_selects`, `Main.postulates_satisfiable`.) The bounds use exact rational arithmetic and mathlib's proved bounds on π. No continuous parameter enters.

a. Premises. The theorem rests on the postulates and on PM's definitions of the pillars. Where a step above reads a PM definition, that reading is stated: Governor restricts transitions and Protocol coordinates them; a vertex sees only local observations (Postulate I); a defect's description consists of the fields listed in Paragraph IV E 0 b; its only invisible symmetry is orientation reversal. Everything else is proved.

b. Causal coupling. Define $\kappa = 1/Z_{\text{geo}}$, the substrate's net admittance per cycle. Lean proves $0<\kappa<1$ (`causalCoupling_bounds`).

VI. EMPIRICAL CORRESPONDENCES

Z_{geo} is a dimensionless property of the substrate. To compare it with experiment we must say which observable it is. This section is outside the main theorem: it adds one physical hypothesis and follows its consequences.

A. Electromagnetic Matching

Suppose coarse graining the substrate yields a local $U(1)$ gauge theory. In rationalized natural units, with the connection normalized so that a unit charge enters as $D_\mu=\partial_\mu+iA_\mu$, write the leading term as $-\frac{1}{4g^2}F_{\mu\nu}F^{\mu\nu}$; then $\alpha=g^2/4\pi$[16].

Matching hypothesis. The substrate's net impedance is the zero-momentum Maxwell coefficient:

$$\frac{4\pi}{g^2(0)} = Z_{\text{geo}}, \qquad \text{so} \qquad \alpha^{-1}(0)=Z_{\text{geo}}, \quad \alpha(0)=\kappa. \tag{19}$$

The quiescent baseline has no external flux and no running fields, so it corresponds to the infrared limit $Q^2\to 0$. The algebra and the inherited bounds are checked in Lean (`electromagnetic_matching`, `physical_inverse_coupling_bounds`). The coefficient 4π is part of the hypothesis: had the calculation produced $Z_{\text{geo}}=C/g^2(0)$, the prediction would be $\alpha^{-1}(0)=(4\pi/C)Z_{\text{geo}}$ (`matching_coefficient`). Deriving Equation (19) requires the substrate's coupling to a background $U(1)$ connection and its quadratic response, which this paper does not compute.

B. The Fine-Structure Constant

Under the matching hypothesis the index predicts $\alpha^{-1}(0)$ with no adjustable parameter (Table IV). It agrees with the rubidium recoil measurement of Morel et al. within $0.58\,\sigma$ and with CODATA 2022 within $1.68\,\sigma$.

Source	$\alpha^{-1}(0)$	$Z_{\mathrm{geo}} - \alpha^{-1}$
Morel (2020), Rb recoil	$137.035\,999\,206 \pm 0.000\,000\,011$ [17]	$+0.58\,\sigma$
Fan (2023), electron $g-2$	$137.035\,999\,166 \pm 0.000\,000\,015$ [18]	$+3.09\,\sigma$
Parker (2018), Cs recoil	$137.035\,999\,046 \pm 0.000\,000\,027$ [19]	$+6.16\,\sigma$
CODATA (2022)	$137.035\,999\,177 \pm 0.000\,000\,021$ [20]	$+1.68\,\sigma$

Table IV. Z_{geo} against determinations of $\alpha^{-1}(0)$. Deviations use the certified value of Z_{geo} and each source's quoted uncertainty (Lean: `alpha_deviations`).

The determinations themselves disagree: the Parker cesium recoil and Morel rubidium recoil values differ by more than $5\,\sigma$, and CODATA's uncertainty reflects that tension.

a. Quantization Gap hypothesis. The substrate has finite channel capacity, while perturbative QED sums ever larger sets of diagrams. We conjecture that high-loop extractions of α, such as from the electron $g-2$, deviate systematically from the kinematic baseline because the continuum expansion eventually exceeds the substrate's capacity. The Fan $g-2$ value lies $3.1\,\sigma$ below the index while Morel agrees within $0.6\,\sigma$, which is the pattern the hypothesis expects. It is not yet a prediction: that requires a mapping from diagrams to channel states, an onset loop order, and the sign and size of the correction to a named observable. Until then the table tests the matching hypothesis, not the gap.

b. Falsification. A robust determination of $\alpha^{-1}(0)$ several standard deviations from Z_{geo}, with its theoretical corrections under control, would refute the conjunction of the postulates and the matching hypothesis.

C. Elementary Charge, Resistance, and Field Scales

The matching hypothesis carries over to the observer's unit system through standard identities. With the Planck charge $q_P = \sqrt{4\pi\epsilon_0\hbar c}$,

$$e = \frac{q_P}{\sqrt{Z_{\mathrm{geo}}}}, \qquad \frac{e}{q_P} = \sqrt{\kappa}. \tag{20}$$

With the impedance of free space $Z_0 = 1/(\epsilon_0 c)$, the von Klitzing constant is

$$R_K = \frac{h}{e^2} = \frac{Z_0}{2}\, Z_{\mathrm{geo}} \tag{21}$$

(`resistance_identity`): the quantum of Hall resistance is half the impedance of free space times the substrate's net impedance per cycle. With the reduced Compton wavelength $\bar{\lambda} = \hbar/m_e c$ and the Schwinger field $E_c = m_e^2 c^3/e\hbar$[21],

$$\frac{e}{4\pi\epsilon_0 \bar{\lambda}^2} = \frac{E_c}{Z_{\mathrm{geo}}} \tag{22}$$

(`schwinger_scale_identity`). These follow from the one hypothesis $\alpha^{-1}(0) = Z_{\mathrm{geo}}$ and are not independent confirmations of it. In the 2019 SI, e and h are exact, so they constrain the relation between α and the measured ϵ_0 (equivalently μ_0)[20]. Charge quantization, the Hall plateaus[22, 23], and pair-production dynamics need further structure.

VII. CONCLUSION

We applied Persistence Mechanics to the vacuum as a closed system and asked whether the requirements of persistence determine its structure. Written as explicit postulates, they do. Finiteness, readability of the substrate's own distinctions, bit-addressed glue, local exchange, and economy admit exactly one substrate up to isometry, the E_8 lattice. Its sectors, channels, signature, boundary, and clock then fix every integer that enters the cost of persistence: $D = 4$, $\Delta = 43$, $\nu = 16$, $N = 32$, $\sigma = 5$, $\chi = 2$. Reading the six pillars as expected transitions per cycle gives the index $Z_{\mathrm{geo}} = 137.035999212\ldots$ with no continuous parameter. Lean checks every step from the postulates to the certified value, using only its standard axioms.

What is proved is an implication. Its premises are the postulates and PM's definitions of the pillars, each stated where it is used, and the postulates are shown to be consistent and independent. Under the further hypothesis that the substrate's net impedance is the infrared Maxwell coefficient, $Z_{\mathrm{geo}} = \alpha^{-1}(0)$, in agreement with the kinematic determination of Morel et al. within $0.58\,\sigma$ and CODATA 2022 within $1.68\,\sigma$.

Three tasks follow. First, derive the matching condition by computing the substrate's response to a background $U(1)$ field. Second, give the Quantization Gap a quantitative form: a correction to a named high-loop observable, with sign and size. Third, build on the same unadjusted substrate the dynamics it was not asked to supply here: a chiral matter spectrum consistent with the known embedding obstructions, a continuum limit, and the remaining constants of the Standard Model. Each is a test the substrate can fail.

If these results hold, physical constants are not free parameters but properties of the least costly structure that can persist on its own. By PM's selection principle, any persistent system solves the same architectural problem; the vacuum is the case with nothing outside it.

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Appendix A: Chirality: Remaining Dynamical Requirements

The D_4 channel bookkeeping does not yet construct the observed chiral fermions. Distler and Garibaldi analyze candidate embeddings into real and complex forms of E_8 and obtain representation-theoretic obstructions to the required chiral gauge theory[9]. Their result is not restricted to an everywhere Euclidean signature. Assigning an effective Lorentzian metric or starting from a discrete lattice does not by itself demonstrate an evasion of their hypotheses.

In particular, an internal Euclidean geometry does not imply absence of a Hamiltonian: internal degrees of freedom can evolve with respect to the spacetime time coordinate. Treating some channels as static at quiescent equilibrium is an approximation or model restriction, not a proof that mirror states have zero propagating capacity.

A physical derivation must specify the microscopic state space and update, derive a low-energy fermion operator, identify its gauge representations and chirality, and explain the fate of any mirror modes while preserving the required conservation laws. It must then state precisely which hypotheses of the embedding obstruction apply. These dynamical tasks remain open and are not implied by the present Lean results.

Appendix B: Geometric Identifiability and the Class Number

Fix an imaginary quadratic order $\mathcal{O}$ and let $\mathcal{T}_{\mathcal{O}}$ denote the nonempty set of complex-isomorphism classes of elliptic curves with endomorphism ring isomorphic to $\mathcal{O}$. The standard CM classification gives a correspondence with proper invertible ideal classes and hence $|\mathcal{T}_{\mathcal{O}}| = h(\mathcal{O})$[14, 24]. This classification is an external mathematical input.

Theorem 8 (Identifiability from ring data alone)**.** *If the observation records only the fixed endomorphism-ring data, then it identifies an element of $\mathcal{T}_{\mathcal{O}}$ uniquely if and only if $h(\mathcal{O}) = 1$.*

Proof. The observation map is constant on $\mathcal{T}_{\mathcal{O}}$. A constant map is injective exactly when its domain has at most one element. The domain is nonempty, so this is equivalent to having exactly one element, which the classification identifies with $h(\mathcal{O}) = 1$. □

Lean checks the abstract constant-map statement as `constant_observation_iff`. For the clock, Lean does not need the general correspondence: in the window $32 \leq |D| \leq 64$ it proves by Gauss reduction that all positive forms of discriminant D are equivalent exactly when $D = -43$ (`clock_unique_iff`).

To reconstruct a physical history one must additionally specify a state space, update law, initial or final data, elapsed time, and an observation map adequate to distinguish the relevant states. A unique torus can support many trajectories. Conversely, reversible evolution is possible within each of several nonisomorphic tori. Class number one is therefore the statement that the clock's geometry is determined by its data, Postulate C; it is distinct from reversibility of the update or reconstruction of a history, which are separate requirements.